Clinical Risk Assessment Report: Patient 21

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 15.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.69x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Patient Age (Age), which reduced the patient's absolute risk by 2.2%. Biologically, this feature patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.18 [Avg (58th %ile)] (+2.1%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.26 [Avg (35th %ile)] (+1.0%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.44 [Avg (43th %ile)] (+0.7%)**

Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 69.0 [Avg (73th %ile)] (-2.2%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.48 [Avg (58th %ile)] (-1.2%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 1.4 [Top 7%] (-1.1%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

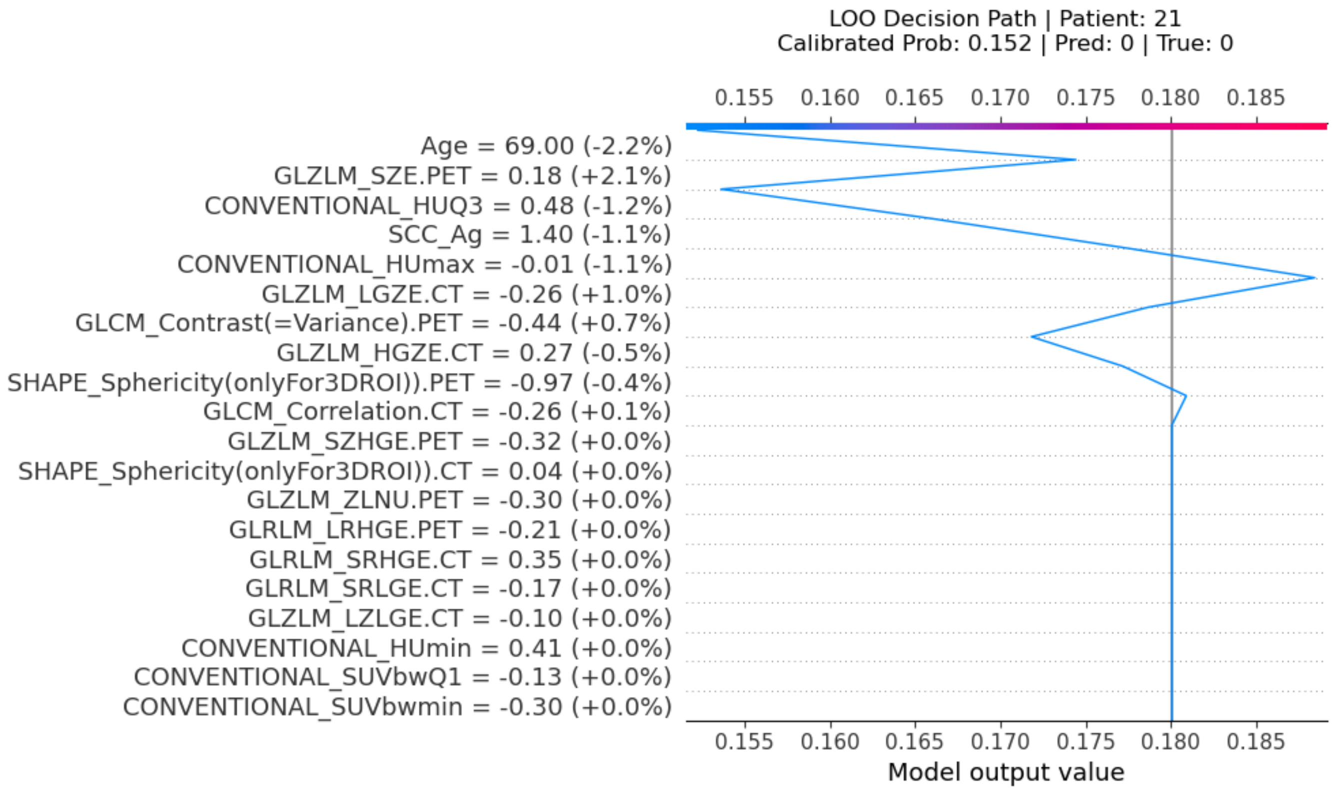
Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.01 [Avg (44th %ile)] (-1.1%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 0.27 [Avg (52th %ile)] (-0.5%)**

Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.